English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, Groups 4/5, 22.01.2022 Test 2

Your grade is 2 plus the number of points earned. Time allowed: 3 hours. Good luck.

Throughout, we work on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$*, and* X, Y *are random variables.*

If needed, you may use $\int e^{-x}\,\mathrm{d}x = -e^{-x}$*,* $\int xe^{-x}\,\mathrm{d}x = -e^{-x}(x+1)$*,* $\int x^2e^{-x}\,\mathrm{d}x = -e^{-x}(x^2+2x+2)$*, and* $\int x^3e^{-x}\,\mathrm{d}x = -e^{-x}(x^3+3x^2+6x+6)$*.*

Problem 1. Human height, as an outcome of natural processes, can be modelled well by a normal distribution. The data used here give a mean height of 163 cm and standard deviation 5.6 cm for women in Bulgaria.

Choose a random group of 20 women.

1. (0.5 points) What is the probability that their average height exceeds 165 cm?
2. (0.5 points) What is the probability that the group includes someone taller than 180 cm?

Problem 2. Let X, Y have joint density $f_{X,Y}(x,y) = ce^{-y}$ for $0 < x < y < \infty$, and zero elsewhere, where c is a constant.

- (0.75 points) Find c and $Cor(X,Y)$.
- (0.25 points) Find $\mathbb{E}(X|Y=1)$.

Problem 3. (1 point)

An enthusiast wants to estimate π using a computer without internet access. They simulate many uniformly distributed points in $[0,1]\times[0,1]$ and examine the proportion falling inside the circle inscribed in the square.

Explain how and why this estimates π. How many points must be simulated so that the probability of an error below 0.001 is 95%?

Problem 4. Let X, Y be independent, with $X \sim U(0,1)$ and $Y \sim Exp(1)$.

1. (0.25 points) Is there an a such that $\mathbb{P}(X \le a) = \mathbb{P}(Y \ge a)$? Does this hold for arbitrary continuous X, Y?
2. (0.75 points) Find the densities of X/Y and Y/X. *(Bonus: Compare their expectations.)*